Live Webinars — Faculty Guide · Modules 01–04 & 07

PCCN Certificate Course · IAP Intensive Care Chapter

Each module has one live webinar. It is **Step 2** of the module pathway: pre-test → recorded lectures → **live webinar** → virtual simulation → module quiz → hands-on workshop.

The webinar is **not a repeat of the lectures**. Delegates arrive having watched the videos. The webinar turns that knowledge into bedside decisions through cases, polls, breakout rooms and drills.

Category	Meaning
Must follow	Rules for every webinar.
Must cover	Content each webinar must work through. The OSCE and final MCQ draw on it.
Better to follow	Recommended facilitation style.
Good to know	Optional, if time allows.

1. Must follow — every webinar

1. **Length:** 60–90 minutes (as set per module below). Start on time and end on time.
2. **Platform:** Zoom, with breakout rooms enabled, polls set up in advance, recording to the cloud, and the waiting room on.
3. **Team for each webinar:**

Role	Responsibilities
Moderator	Runs the clock, introduces faculty, manages chat questions, closes the session.
Lead faculty	Teaches the cases and the drills.
Breakout facilitators	1 for every 8–10 delegates; each has the case card and the answer key.
Tech host	Polls, breakout rooms, screen-share backup, recording, attendance export.

4. **Attendance:**
   - Delegates need **70% attendance** for Step 2.
   - Export the Zoom attendance report within 24 hours and send it to the course coordinator.
   - Delegates who miss the session watch the recording (the recorded makeup) and complete the 5-question check in the portal.
5. **Interaction:** at least **one poll every 10 minutes**, one breakout activity, and one “pause and decide” case. Delegates are on camera during breakouts where bandwidth allows.
6. **Content rules (same as the lectures):**
   - Current AHA PALS, APLS and IAP guidance.
   - Doses per kg with the maximum, generic names only, matching the course dose tables.
   - British spelling, SI units.
   - No commercial content.
7. **Privacy:** no patient names, faces or hospital IDs. Real case clips need written consent and blurred faces. Delegates are reminded not to share patient photos in the chat.

8. **Psychological safety:** open with the basic assumption (everyone is intelligent, capable and wants to improve). Correct errors kindly and without naming the person.
9. **DIRECT throughout:** every case is worked through Detection, Intervention, Reassessment, Effective communication and Transport.
10. **Deliverables after the webinar (within 48 hours):**
    - Recording link to the portal team.
    - Attendance report.
    - Poll results.
    - The 5 most important questions from the chat, with short written answers for the FAQ.

2. Must follow — running order of every webinar

Time	Segment
0:00–0:05	Welcome, ground rules, basic assumption, objectives (3)
0:05–0:10	**Warm-up poll** — 3 questions from the module pre-test
0:10–0:25	**Case 1, run live** (lead faculty), with a poll at each DIRECT step
0:25–0:50	**Breakout rooms** — the module drill (see the module brief), facilitators with the case card
0:50–1:05	**Report back** — one group per room presents; faculty correct and consolidate
1:05–1:15	**Case 2 or skills demonstration**
1:15–1:25	Q&A from the chat
1:25–1:30	**3 take-home points**, pointer to the virtual simulation and the quiz, feedback form

For 60-minute webinars, shorten the breakout to 15 minutes and drop Case 2.

3. Better to follow — facilitation style

- **Camera on and talk to the nurses.** Put the bedside view first.
- **Cold-call kindly:** “Sister Priya, you are at the bedside — what do you do first?”
- **Use the chat waterfall:** everyone types their answer, but only presses Enter on the moderator’s signal. This avoids anchoring on the first answer.
- **Show real equipment on camera:** the crash cart, an HFNC circuit, a nebuliser, a pump, an EVD set.
- **Close every case** with “What would you say in your ISBAR call?”

4. Module-by-module brief

Module 01 · Initial Assessment — “PICU Layout & Emergency Preparedness” (90 min)

Must cover

- **Crash cart organisation:**
    - Drawers in order: Airway, Breathing, Circulation/access, Drugs, Defib/ECG, Miscellaneous.
    - Daily seal and expiry checks; restock straight after every code.
- **The 7 P’s of emergency preparedness:** Staff, System, Supplies, Space, Speak up, Subject, Self.
- **SOAPME** before every procedure.

- **In-hospital triage levels 1–5.** Mass-casualty and pandemic surge triage only as an introduction.
- **Breakout drill:** each room audits a photo of a crash cart (provided) and lists 5 errors. Then each room triages 6 children waiting in the emergency room using PAT and ABCDE.

Good to know: the WHO surgical safety checklist evidence; the Hudson River case as a lesson in preparation.

Module 02 · Respiratory Support — "Stridor & Wheeze; NIV & the Tracheostomised Child" (90 min)

Must cover

- **Bedside differentiation** of upper airway disease (stridor) from lower airway disease (wheeze). Escalation triggers and when to call for help.
- **NIV:** how it helps (lower work of breathing, higher FRC, correction of hypercarbia) and why it fails, mask discomfort being the commonest cause.
- **Sedation for NIV** at the course dose: dexmedetomidine 0.3–1 mcg/kg/h, on its own line and never bolused.
- **Tracheostomy care:** stoma care, humidification, suctioning depth, and **emergency tube change** demonstrated live.
- **Breakout drill:** the "Escalation Drill" case. Each step (nasal prongs → HFNC → bag-mask) is a group decision with an ISBAR call.

Good to know: heliox; tracheostomy emergency algorithms (NTSP).

Module 03 · Shock & Haemodynamics — "The Septic Child + Fluid Calculation Workshop" (90 min)

Must cover

- **The 1-hour sepsis bundle**, walked through live:
  - Cultures before antibiotics.
  - Antibiotic (ceftriaxone 100 mg/kg, max 4 g).
  - Fluid bolus by push–pull.
  - Vasoactive drug if fluid-refractory.
  - Stress-dose hydrocortisone only if the child is at risk (1–2 mg/kg).
- **Calculation workshop:**
  - Maintenance by the 4-2-1 rule, restricted to 50–70% while the child is sick.
  - Deficit and replacement.
  - Bolus volumes per kg.
  - Peripheral adrenaline/noradrenaline infusion preparation and rate.
- **Breakout drill:** each room calculates the fluids and the vasoactive infusion for a 20 kg child. A second nurse checks every calculation, and rooms swap answers to check each other.

Good to know: ROSE fluid phases; the 4 Ps before vasoactive drugs.

Module 04 · Arrhythmias, BLS & Resuscitation — "Mega-Code Debrief" (90 min)

Must cover

- **Running a code as a team:**
  - A named team leader.
  - Clear roles: compressor, airway, drugs/cart, defibrillator, recorder.
  - Closed-loop communication.
  - Swap compressors every 2 minutes.
- **The defibrillator under time pressure:**

    - Shockable versus non-shockable rhythms.
    - Energy 2 J/kg, then 4 J/kg (maximum 10 J/kg or the adult dose).
    - Pad placement and a safe shock.
- **Emergency drugs at course doses:**
    - Adrenaline 0.01 mg/kg (0.1 mL/kg of 1:10 000), max 1 mg.
    - Amiodarone 5 mg/kg (max 300 mg).
    - Adenosine 0.1 mg/kg (max 6 mg), then 0.2 mg/kg (max 12 mg).
    - Atropine 0.02 mg/kg (max 0.5 mg).
- **Breakout drill:** a timed mega-code on paper. Rooms call each action aloud with timestamps, and the facilitator scores time to first shock and pauses in CPR.

Good to know: post-cardiac-surgery JET and complete heart block; post-ROSC care.

Module 07 · Neurological Emergencies — "Neuro Night Shift: GCS Calibration & the Seizure Clock" (90 min)

Must cover

- **GCS calibration (0:10–0:25):**
    - Three short consented clips (infant, toddler, school-age child). Every delegate scores E, V and M in a poll.
    - Faculty reveal the agreed score, and the group discusses the common errors: verbal scoring in pre-verbal children, and best motor response versus withdrawal.
- **Seizure clock, run live (Case 1):**
    - A 3-year-old is still seizing at 15 minutes after diazepam given by the GP.
    - The moderator runs a visible timer. Delegates answer at each minute mark:
        - **0–5 min:** stabilise and check glucose.
        - **5–10 min:** benzodiazepine — **the second and last dose**.
        - **20 min:** second-line drug, with the dose calculated live.
        - **40 min:** refractory — call the PICU, prepare with SOAPME.
- **Breakout drill — "10 commandments" cluster-care plan:**
    - Each room receives a 7-year-old with meningitis and raised ICP.
    - Each room writes the first 4 hours of nursing care, covering: position, cluster care, sedation, glucose, sodium, oxygenation, CO_2, temperature and seizures, hyperosmolar therapy, and neuro-observation frequency.
    - The room also writes the trigger for calling the doctor (Cushing's triad, unequal pupils).
- **Report back:** one plan per room. Faculty mark against the 10 commandments, and every room fixes one gap.
- **Case 2 / demonstration:**
    - EVD levelling to the tragus on camera; when to clamp.
    - What to chart hourly and when to call.
    - Shift and drift.
- **Communication:**
    - An ISBAR call for a falling GCS, in which a delegate role-plays the call.
    - How to talk to parents during a prolonged seizure.

Good to know: non-convulsive status; the acute encephalitis syndrome (AES) and Japanese encephalitis (JE) programmes in India.

Materials the faculty prepare:

- 3 GCS clips (consented and blurred).
- A seizure-clock slide with a timer.
- A breakout case card with an answer key.
- An EVD set or a photo of one.
- A dose sheet matching the Module 07 dose table.

5. Faculty preparation checklist (one week before)

- ☐ Read the module's lectures and simulation so the webinar builds on them
- ☐ Case cards and answer keys sent to the breakout facilitators
- ☐ Polls (at least 6) sent to the tech host
- ☐ Slides in the PCCN template; doses checked against the course dose table
- ☐ Clips consented, blurred, and tested on screen share
- ☐ 15-minute tech rehearsal the day before (screen share, polls, breakouts, recording)
- ☐ Feedback form link ready for the close

6. Schedule

Module	Webinar	Lead faculty	Moderator	Facilitators	Date / time
01	PICU Layout & Emergency Preparedness				
02	Stridor & Wheeze; NIV & Tracheostomy				
03	The Septic Child + Fluid Calculation				
04	Mega-Code Debrief				
07	Neuro Night Shift: GCS & the Seizure Clock				

References: AHA PALS 2020; APLS 7th ed (2023); Surviving Sepsis Campaign Paediatric Guidelines 2020; ILAE 2015 definition of status epilepticus; AES Guideline 2016; Paediatric Severe TBI Guidelines 2019; PCCN Course Compendium Sections 1–5 (Section 5 = portal Module 07).